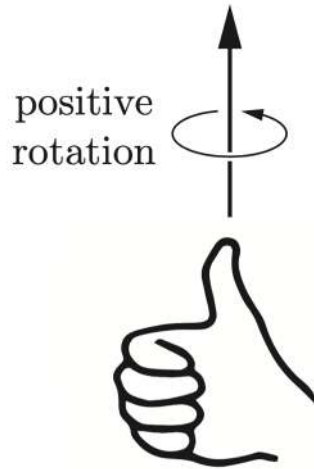
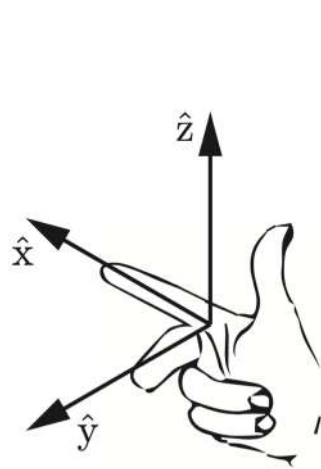


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
	3.2 Rotations and Angular Velocities
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

- We often define a fixed **space frame** $\{s\}$ and a **body frame** $\{b\}$ attached to some body of interest. All frames are *instantaneously stationary*.
- Right-handed frames, and right-hand rule for positive rotation.



orthogonal

special

- **Special orthogonal group** $SO(3)$: matrices R in $\mathbb{R}^{3 \times 3}$ where $R^T R = I$, $\det R = 1$. R is a **rotation matrix**. Implicit representation with 9 numbers for 3 dof.

Important concepts, symbols, and equations (cont.)

- A **group** is a set of elements $G = \{a, b, c \dots\}$ and a binary operation $\bullet$ satisfying

closure

$$a \bullet b \in G \text{ for all } a, b \in G$$

associativity

$$(a \bullet b) \bullet c = a \bullet (b \bullet c)$$

identity element exists

there is an $I \in G$ such that $a \bullet I = I \bullet a = a$
for each $a \in G$

inverse exists

for each $a \in G$, there exists $a^{-1} \in G$ such that
 $a \bullet a^{-1} = a^{-1} \bullet a = I$

Integers under addition? Nonnegative integers under addition? Square real matrices under multiplication?
What is a **Lie group**?

Important concepts, symbols, and equations (cont.)

- $SO(3)$ is a **matrix (Lie) group** (the group operation is matrix multiplication).

closure: $R_1 R_2 \in SO(3)$

associative: $(R_1 R_2) R_3 = R_1 (R_2 R_3)$ (*not commutative!* $R_1 R_2 \neq R_2 R_1$ generally)

identity: identity matrix I

inverse: matrix inverse

$$R^T R = I, \text{ so } R^{-1} = R^T.$$

$$\text{For } x \in \mathbb{R}^3, \|x\| = \|Rx\|.$$

Important concepts, symbols, and equations (cont.)

- Uses of a rotation matrix:
 1. Represent an orientation. R_{ab} represents orientation of $\{b\}$ in $\{a\}$.
 2. Change the reference frame of a vector or frame.

subscript cancellation:

$$R_{ab}R_{bc} = R_{a\cancel{b}}R_{\cancel{b}c} = R_{ac}$$

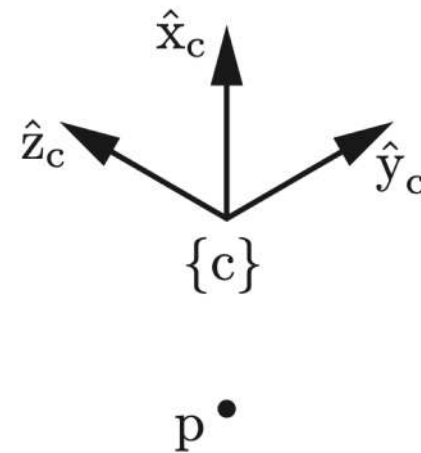
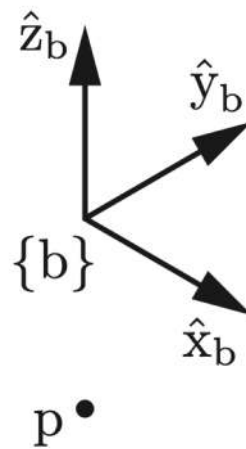
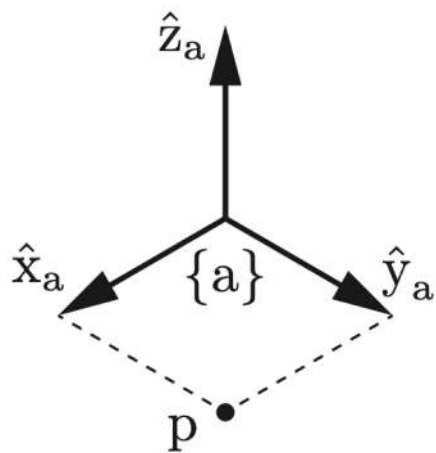
$$R_{ab}p_b = R_{a\cancel{b}}p_{\cancel{b}} = p_a$$

3. Rotate a vector or frame. $R = R_{cd} = \text{Rot}(\hat{w}, \theta)$, axis $\hat{w}$ expressed in $\{c\}$.

$$p'_c = R_{cd}p_c \quad (\text{no subscript cancellation})$$

$$R_{ab}' = RR_{ab} \quad (\text{after rotating about axis in } \{a\})$$

$$R_{ab}'' = R_{ab}R \quad (\text{after rotating about axis in } \{b\})$$



$$R_{ab} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$p_b =$$

$$p_a = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

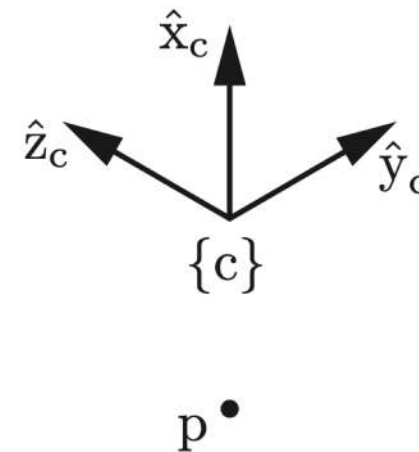
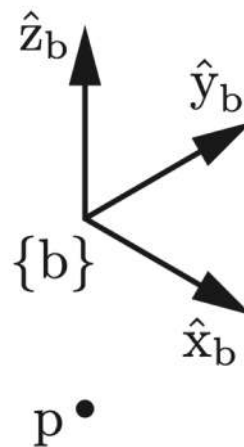
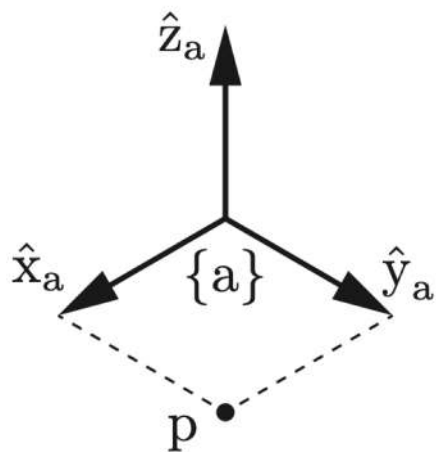
$$R_{ba} p_a = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

Given $R_1 = R_{ab}$, $R_2 = R_{bc}$, and $R_3 = R_{ad}$,
write R_{dc} in terms of R_1 , R_2 , and R_3 (no inverses!).

$$\begin{aligned} R_{dc} &= R_{da} R_{ab} R_{bc} \\ &= R_3^T R_1 R_2 \end{aligned}$$

Given p_b , what is p_d in terms of R_1 , R_2 , and R_3 (no inverses)?

$$\begin{aligned} p_d &= R_{da} R_{ab} p_b \\ &= R_3^T R_1 p_b \end{aligned}$$



$$R = R_{ba} = \text{Rot}(\hat{w}, \theta): \theta = \pi/2, \text{ axis } \hat{w} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$$R_{bc}' = RR_{bc} = \text{rotate } \{c\} \text{ by } \pi/2 \text{ about } -\hat{z}_b$$

$$R_{bc}'' = R_{bc}R = \text{rotate } \{c\} \text{ by } \pi/2 \text{ about } -\hat{z}_c$$

orientation representation	# nums	imp/exp?	pros	cons
Euler angles, roll-pitch-yaw	3	explicit	minimum #s	singularities (gimbal lock)
Unit quaternions S^3 covers rotations twice	4	implicit	no singularities fewer numbers than $so(3)$ easy to interpolate easy to normalize back to S^3	unintuitive
Rotation matrices	9	implicit	linear algebra linear diffeqs integrate angular velocities unique R for each orientation	lots of numbers more computations to snap back to $so(3)$